Efe Cagin Yildirim

Rennes, France | efecaginyildirim@gmail.com
linkedin.com/in/efecaginyildirim | github.com/EfeCaginYildirim

Seeking a 10 to 12-week engineering internship from July 2027 – embedded systems, electronics or data science

Education

INSA Rennes — Rennes, France
Diplôme d'Ingénieur (Master's level) – Electronics and Industrial Computing (4th year) — *2022 – 2028*

Politecnico di Torino — Turin, Italy
Erasmus+ Exchange – Electronic Engineering — *02/2027 – 06/2027*

- Planned courses: Analog and Telecommunication Electronics, Radar and Remote Sensing, Operating Systems

Galatasaray High School — Istanbul, Turkey
High School Diploma, science track – French-Turkish bilingual program (competitive entrance exam) — *2018 – 2022*

Professional Experience

Otokar | *Java, SQL Server, JDBC, REST, JUnit, Maven* — Sakarya, Turkey
Intern – Digital and Smart Manufacturing Technologies — *07/2025 – 09/2025*

- European MATISSE project (Digital Twin): developed an automated collector for sensor data from motors
- Parallel collection through a REST API (thread and HTTP connection pools, automatic retry on server errors), daily scheduling and incremental retrieval of new data only
- Batch insertion into SQL Server with `MERGE` (5,000 rows per batch), composite-key deduplication and timestamp validation; 50 JUnit unit tests

Lucra Marble — Denizli, Turkey
Production Intern – quality control on a marble production line — *06/2023*

- Quality inspection of marble slabs and observation of the line; presented findings to supervisors

Projects

IBM Capstone – Falcon 9 Landing Prediction | *Python, pandas, SQL, scikit-learn, Plotly Dash* — 08/2026

- End-to-end pipeline on 90 SpaceX launches: data collection via REST API and web scraping (BeautifulSoup), pandas data cleaning, 10 SQL queries, interactive Folium map and Plotly Dash dashboard
- 4 classifiers (logistic regression, SVM, decision tree, KNN) tuned with GridSearchCV: 87.5% cross-validation accuracy, 83.3% test accuracy

Real-Time Digital Oscilloscope | *C, STM32, GTK+3, SDL2, UART* — 01/2026 – 06/2026

- Acquisition on an STM32 microcontroller at 500 kHz (12-bit ADC, 0.37 mV resolution) streamed over UART; lock-free multithreaded MVC (Model-View-Controller) architecture with a 10M-sample circular buffer
- Responsible for the GTK+3 user interface and the measurement cursors (Δt, ΔV); 100% test coverage on the Model

Transistor Curve Tracer | *Analog electronics, OrCAD PSpice* — 05/2026 – 06/2026

- Board plotting $I_C = f(V_{CE})$ of NPN transistors on an oscilloscope (XY mode): 0–18 V ramp from a non-inverting amplifier, 4-level current source (5 to 40 μA) and INA139 current sensing, validated in PSpice simulation

Ultrasonic Obstacle Detection Receiver | *Analog electronics, OrCAD PSpice* — 12/2025 – 01/2026

- 40 kHz receive chain detecting obstacles up to 1 m: transistor amplifier with adjustable gain (5–40), precision rectifier and active envelope filter; 77.5 mW power consumption

CPU Execution Unit Extension | *Digital design, RTL* — 12/2025 – 01/2026

- Iterative divider controlled by a finite state machine (FSM), 16-bit adder-subtractor, frequency divider

Certifications

IBM Data Science Professional Certificate (IBM / Coursera) – Python, SQL, data analysis, machine learning

Technical Skills

Programming: C, C++, Java, Python, SQL, VHDL, Bash
Embedded Systems: STM32, FPGA, Quartus Prime, ModelSim, UART, Linux
Electronics & Signal Processing: analog electronics, signal processing, filtering, OrCAD, PSpice, KiCad
Data Science & Machine Learning: NumPy, pandas, Matplotlib, Seaborn, scikit-learn, Plotly Dash, Folium, Jupyter
Tools: MATLAB, Simulink, Git, GitLab, Maven, CATIA, Microsoft Office

Languages

Turkish (native) · French (bilingual) · English (fluent) · Italian (intermediate)

Interests

Fitness, football, basketball · Member of the Congress (general assembly) of Galatasaray SK since 2025